

Digital Signal Processing 2026

General requirements for the project report

Minimal required structure

1. Title and authors
2. Aim and research question
3. Background
4. Data or test signals
5. Preprocessing
6. Method
7. Validation or demonstration
8. Results
9. Course concepts& methods used
10. Discussion and limitations
11. Conclusion
12. References

1. Basic information

The report must include:

- project title;
- student names;
- date;

2. Aim and research question

State clearly what the project investigates.

The report should include one or two concrete questions, for example:

How does the selected method work, and what are its main limitations?

How does method A compare with method B on the same signal or task?

Can the method be applied successfully to a real or realistic signal?

3. Background

Briefly explain the theoretical or practical context of the project.

This section should answer:

- What is the method, algorithm, or signal-processing concept?
- Why is it useful?
- What problem does it solve?
- What are its known limitations or trade-offs?

Examples include filtering, Fourier analysis, wavelets, Hilbert transform, spectral estimation, fingerprinting audio signals, peak detection, or time-frequency analysis.

4. Data or test signals

Describe the signals used in the project.

Depending on the project, this may include:

- synthetic signals generated by the students;
- audio recordings;
- EEG or other physiological signals;
- images or other sampled data;
- publicly available datasets;
- self-recorded or provided data.

State clearly:

- sampling frequency or sampling interval;
- signal duration;
- number of channels, if relevant;
- source of the data;
- whether the data were real, synthetic, or both.

5. Preprocessing

Describe all preprocessing steps before the main analysis.

Possible examples:

- resampling;
- normalization;
- filtering;
- segmentation into windows or epochs;
- artefact or noise removal;
- conversion between stereo and mono;
- trimming or selecting signal parts.

Avoid vague descriptions such as:

The signal was cleaned.

Instead, state exactly what was done.

6. Method

Describe the implemented method step by step.

Include:

- the main equations, if relevant;
- parameter values;
- algorithmic steps;
- software functions or libraries used;
- whether the method was manually implemented or package-based;
- justification of important choices.

For example:

- for a filter project: filter type, order, cutoff frequency, frequency response;
- for a Hilbert transform project: analytic signal, envelope, instantaneous phase and frequency;
- for a Shazam-like project: spectrogram, peak detection, fingerprints, matching rule;
- for a wavelet project: mother wavelet, scales/frequencies, number of cycles;
- for a Fourier project: window length, overlap, spectral resolution.

7. Validation or demonstration

Every project must include a clear demonstration that the method works.

This may be done using:

- a synthetic signal with known ground truth;
- a simple example where the expected result is known;
- comparison with another method;
- comparison before and after processing;
- robustness test with noise or changed parameters;
- visual inspection of representative examples.

The report should not only show results; it should show why the results are believable.

8. Results

Present the main results using clear figures and tables.

Figures must have:

- axis labels;
- units;

- readable captions;
- legends where needed;
- comparable scales when methods or parameters are compared.

Tables must have meaningful column names and reasonable numerical precision.

The results should directly answer the research question.

9. Course concepts used

The report must explicitly identify which topics from the DSP course were used in the project.

This section should not be only a list of keywords. For each item, briefly explain how it appeared in the project. For example:

Course topic	How it was used in the project
Fourier transform	Used to compute the spectrum of the signal and identify dominant frequency components.
Windowing	Used in the STFT/Welch method to analyze short signal segments.
Filtering	Used to isolate the frequency band of interest before further analysis.
Convolution	Appears in the implementation of LTI filters and wavelet transforms.
Phase	Considered when comparing causal and zero-phase filtering.
Sampling	Used to define the frequency axis, Nyquist frequency, and valid cutoff frequencies.

Only include the topics that were actually used in the project.

10. Comparison and parameter effects

Where relevant, compare at least two methods, settings, or conditions.

Examples:

- different filter orders;
- different window lengths;
- different wavelet parameters;
- clean versus noisy signals;
- synthetic versus real data;
- manual implementation versus package function;
- original signal versus processed signal.

The report should explain what changed and why.

11. Discussion and limitations

Interpret the results.

Answer:

- Did the method work?
- What did the results show?
- What are the practical implications?
- What are the main limitations?
- What would be improved with more time?

Typical limitations include:

- noise;
- short signals;
- small dataset;
- parameter sensitivity;
- lack of ground truth;
- computational cost;
- use of simplified assumptions.

12. Conclusion

End with a short conclusion.

It should state:

- what was implemented or studied;
- the main result;
- the main limitation or recommendation.

13. Code and reproducibility

Submit the code or notebook together with the report.

The code should:

- run from beginning to end;
- contain a clear configuration section;
- use clear variable names;
- save or display the main figures;
- include comments for important steps.

The report should state:

- programming language;
- main packages used;
- how to run the code;
- what files are needed.

14. References and formatting

The report must cite:

- the assigned paper or main source, if applicable;
- important method references;
- software packages or datasets used.

Recommended length:

6 – 10 pages

excluding appendices.

Use clear section headings, numbered figures, numbered tables, and concise explanations.